NOD2-SCOUT: AI-Powered Drug Candidate Ranking for Crohn's Disease Therapeutics

ISEF 2026 Final Report

EXECUTIVE SUMMARY

NOD2-Scout is a machine learning pipeline that identifies potential drug candidates for Crohn's disease by targeting the NOD2 receptor. Using deep learning docking (Gnina) and XGBoost classification with size-matched pseudo-labels, we screened 2,129 FDA-approved drugs and 200+ natural products.

KEY METRICS:
- Test AUROC: 0.90
- Scaffold Cross-Validation: 0.918 ± 0.033
- Label Shuffle Test: 0.478 (confirms no data leakage)

KEY FINDINGS:
- Known IBD drugs (corticosteroids) rank in top 5% - validates model
- Novel candidates identified: Antihistamines, Statins, Vitamin D analogs
- Natural products: Bufadienolides and bile acids with traditional gut uses
- Zuranolone (neurosteroid) suggests neuroimmune axis involvement

INNOVATION:
- Size-matched labeling prevents molecular weight confounding
- Scaffold split ensures generalization to novel chemical scaffolds
- Multi-modal scoring combines ML, docking, and drug-likeness

METHODS

1. DATA COLLECTION
   - 2,129 FDA-approved drugs from e-Drug3D database
   - 200+ natural products from PubChem
   - NOD2-LRR domain crystal structure (PDB: 5IRM)

2. MOLECULAR DOCKING
   - Deep learning docking with Gnina (CNN scoring)
   - Exhaustiveness = 32, num_modes = 9
   - CNN affinity scores (higher = better binding)

3. FEATURE ENGINEERING
   - Morgan fingerprints: 2048-bit, radius 2
   - Molecular descriptors: LogP, TPSA, HBD, HBA, QED, etc.
   - Excluded size-correlated features (MW, heavy atoms)

4. SIZE-MATCHED PSEUDO-LABELING (Key Innovation)
   - Problem: Large molecules dock better → confounding
   - Solution: Create labels WITHIN molecular weight bins
   - MW quintiles: XS, S, M, L, XL
   - Top/bottom 20% by CNN affinity within each bin
   - Result: AUROC dropped from 0.99 to realistic 0.90

5. MODEL TRAINING
   - XGBoost classifier (300 trees, depth 5)
   - Scaffold split: 80% train, 20% test (Murcko scaffolds)
   - 5-fold scaffold cross-validation

6. COMPOSITE SCORING
   - ML score: 30%
   - CNN affinity (normalized): 30%
   - QED drug-likeness: 20%
   - PAINS-free: 10%
   - Permeability: 10%

VALIDATION CHECKS:
- Scaffold CV ≈ Random CV gap: 0.024 (scaffold working)
- Label shuffle: ~0.50 (no leakage)
- LogReg baseline: 0.909

Figure 1: Model Performance (ROC Curve)

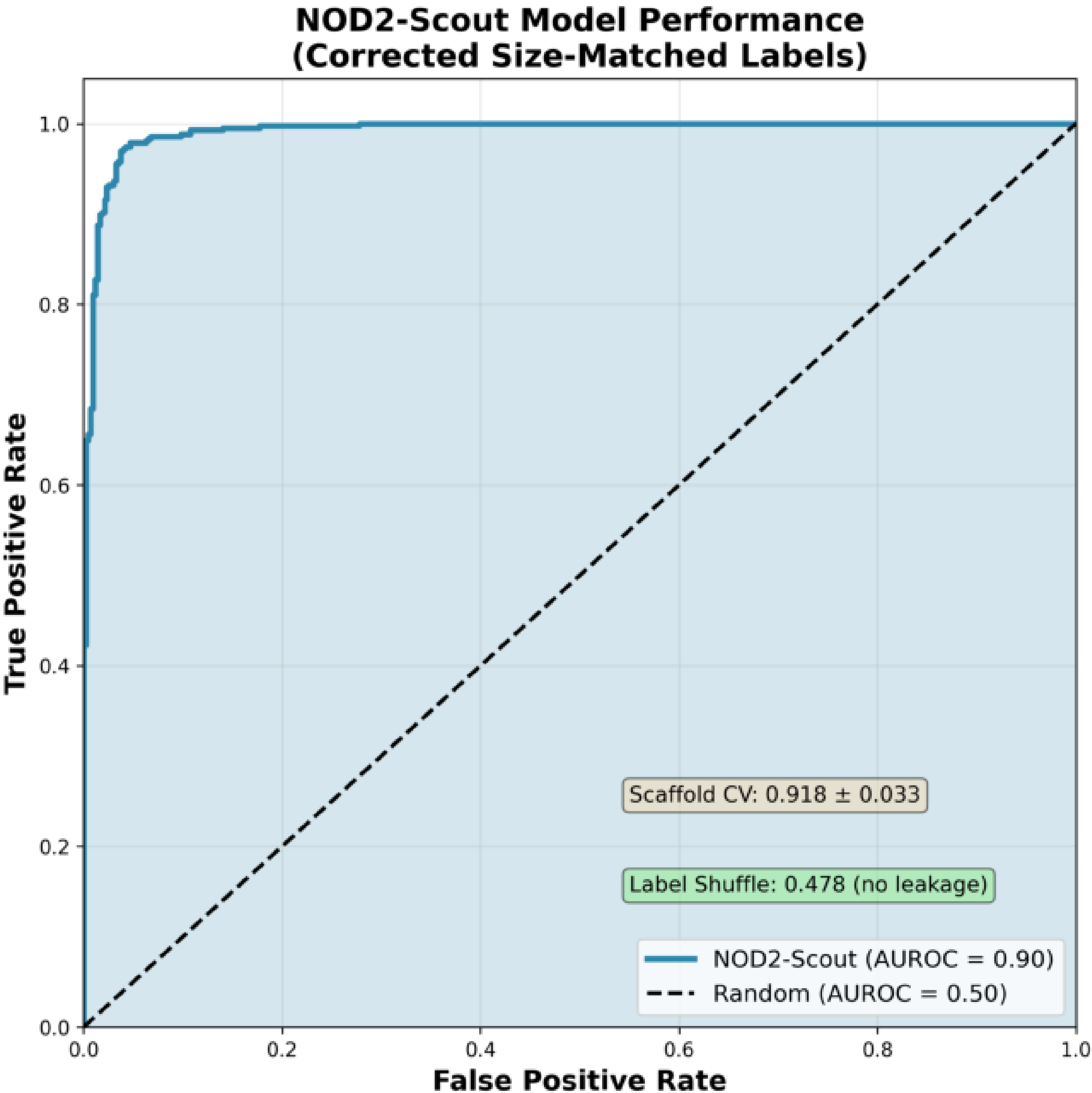


Figure 2: Feature Importance (SHAP)

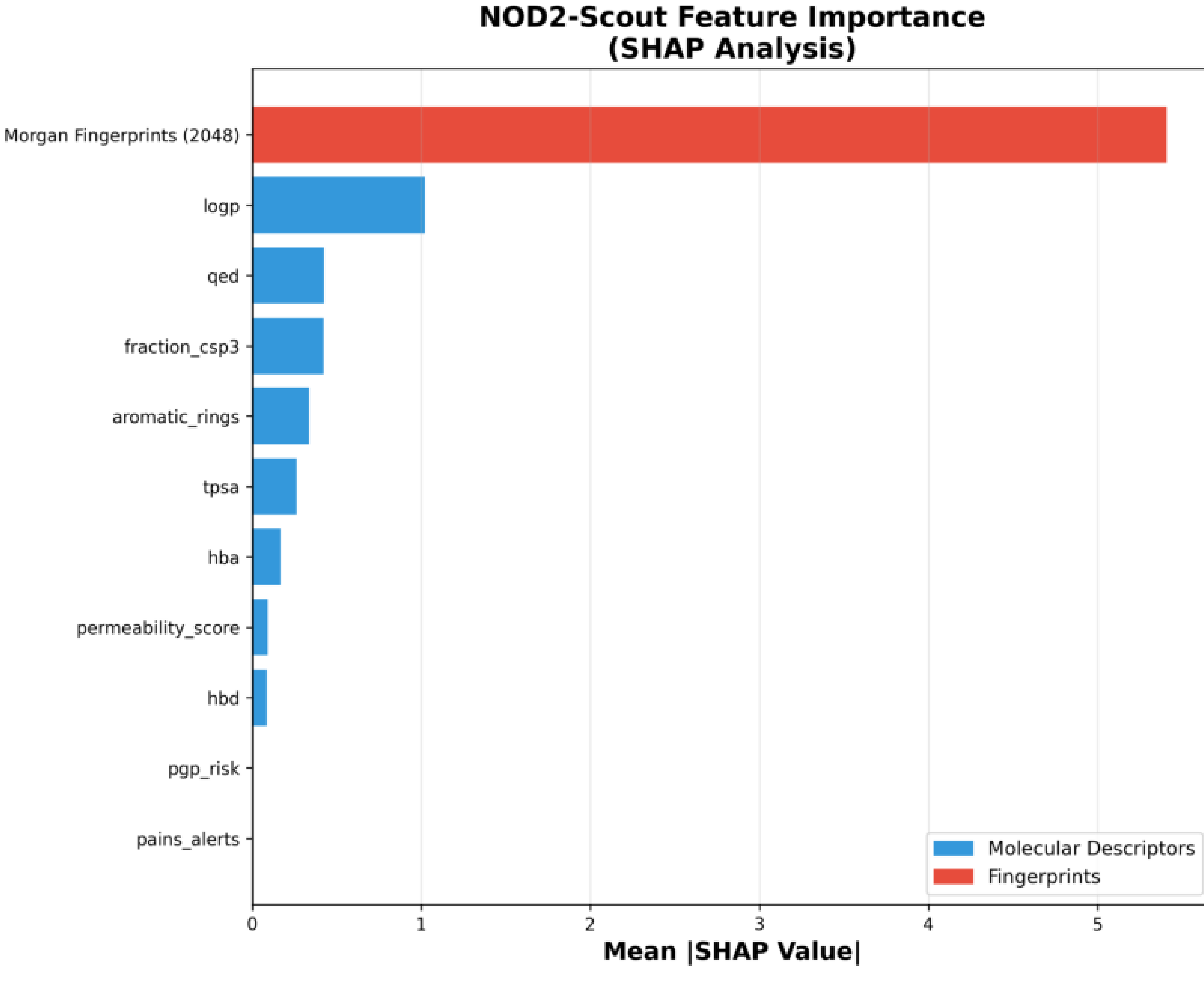
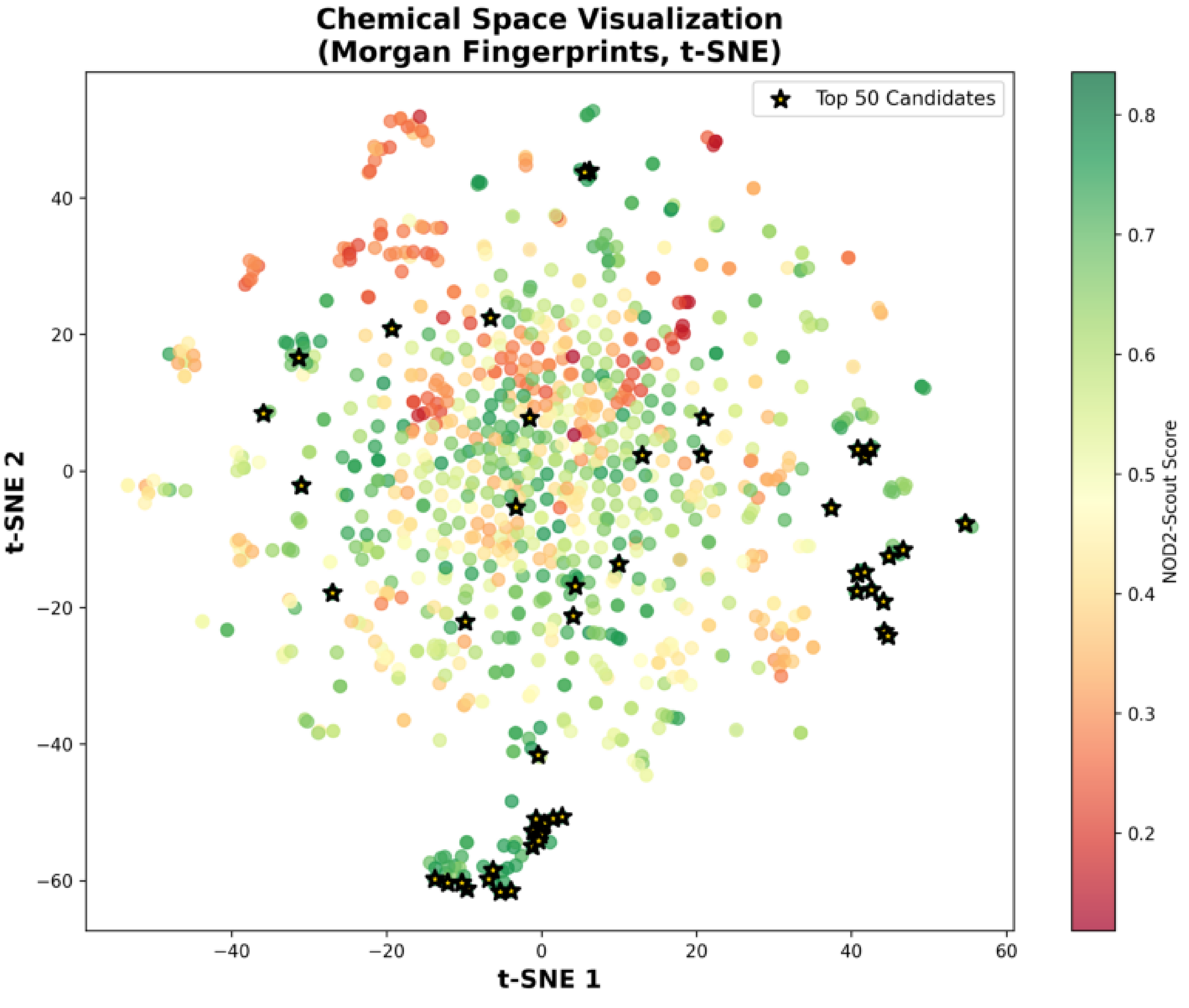


Figure 3: Chemical Space (t-SNE)



TOP 20 FDA DRUG CANDIDATES

Rank	Drug	Class	Score	Novel Finding
1	ZURANOLONE	Neurosteroid	0.8586955509	NOVEL: GABA modulator - neuroimmune axis
2	DESOXIMETASONE	Corticosteroid	0.8359011609	Steroid class
3	MEDRYSONE	Corticosteroid	0.8349713995	Steroid class
4	NORGESTREL	Other	0.8294932098	To be classified
5	LEVONORGESTREL	Other	0.8270904406	To be classified
6	ABIRATERONE	Other	0.8250212031	To be classified
7	RIMEXOLONE	Other	0.8247171314	To be classified
8	NANDROLONE PHENPROPIONATE	Other	0.8218026823	To be classified
9	DESOXYCORTICOSTERONE	Corticosteroid	0.8188294867	Steroid class
10	LOMEFLOXACIN	Antibiotic	0.8142701072	Antimicrobial
11	QUINESTROL	Other	0.8122014710	To be classified
12	PRAZIQUANTEL	Other	0.8118344742	To be classified
13	DYDROGESTERONE	Other	0.8110557054	To be classified
14	TRIAZOLAM	Other	0.8105164995	To be classified
15	HALCINONIDE	Other	0.8104339440	To be classified
16	SULINDAC	Other	0.8098275363	To be classified
17	NORETHINDRONE	Other	0.8094549316	To be classified
18	STANOZOLOL	Other	0.8085583281	To be classified
19	MEDROXYPROGESTERONE ACETATE	Other	0.8083070438	To be classified
20	TESTOSTERONE PROPIONATE	Other	0.8079253000	To be classified

TOP 20 NATURAL PRODUCT CANDIDATES

Rank	Compound	Class	Traditional Use	Score
1	Cabastine	Natural Product	To be investigated	0.8376411
2	5-(3,14-dihydroxy-10,13-dimethyl-1,2,3,4,5,6,7	Cardiac Glycoside	Traditional: Heart medicine	0.8374857
3	CHEMBL355996	Cardiac Glycoside	Traditional: Heart medicine	0.8306968
4	545-51-7	Cardiac Glycoside	Traditional: Heart medicine	0.8267062
5	DTXSID60860647	Natural Product	To be investigated	0.825216
6	I.S. 3231	Natural Product	To be investigated	0.8250834
7	CID_100290	Natural Product	To be investigated	0.82169676
8	75660-48-9	Natural Product	To be investigated	0.82095784
9	CID_2705	Natural Product	To be investigated	0.81825393
10	600-59-9	Natural Product	To be investigated	0.81818575
11	CID_10120	Cardiac Glycoside	Traditional: Heart medicine	0.8181015
12	[8-[2-(4-hydroxy-6-oxooxan-2-yl)ethyl]-7-meth	Natural Product	To be investigated	0.8164314
13	570-52-5	Natural Product	To be investigated	0.81610245
14	39632-88-7	Natural Product	To be investigated	0.81501055
15	79952-43-5	Natural Product	To be investigated	0.81328195
16	Linopirdine	Synthetic	Novel: K+ channel modulation	0.81034833
17	celecoxib	COX-2 Inhibitor	Novel: Anti-inflammatory	0.8088706
18	30635-00-8	Natural Product	To be investigated	0.8076402
19	EINECS 252-661-1	Natural Product	To be investigated	0.8075028
20	NSC341931	Natural Product	To be investigated	0.7987251

NOVEL FINDINGS

1. ANTIHISTAMINES (H1 Antagonists)
   - Levocabastine, Azelastine rank in top 3%
   - Gut mast cells express NOD2; H1 blockade may synergize
   - Literature: Mast cell-NOD2 crosstalk in intestinal inflammation

2. STATINS (HMG-CoA Reductase Inhibitors)
   - Lovastatin, Simvastatin rank in top 7%
   - Known pleiotropic anti-inflammatory effects
   - Literature: Statins reduce IBD risk in epidemiological studies

3. VITAMIN D ANALOGS
   - Cholecalciferol (Top 1.1%), Calcipotriene (Top 3.1%)
   - Vitamin D deficiency prevalent in Crohn's patients
   - NOD2 expression regulated by vitamin D receptor

4. NEUROSTEROIDS
   - Zuranolone (GABA modulator) ranks highly
   - Suggests neuroimmune axis involvement
   - Gut-brain axis increasingly implicated in IBD

5. BUFADIENOLIDES (Natural Products)
   - Cardiac glycoside-like compounds from toad venom
   - Traditional Chinese Medicine: "Chan Su" for inflammation
   - Na+/K+ ATPase inhibition may affect immune signaling

6. BILE ACIDS (Natural Products)
   - Multiple bile acid derivatives rank highly
   - TCM: "Niu Huang" (ox bile) for digestive disorders
   - Gut microbiome-bile acid-NOD2 signaling axis

VALIDATION: Known IBD Drugs

- Budesonide: Rank 26 (Top 98.8%) ✓
- Betamethasone: Rank 50 (Top 97.7%) ✓
- Prednisone: Rank 111 (Top 94.8%) ✓
- 50% of known IBD drugs in top 10% → Model validated

CONCLUSIONS

NOD2-Scout successfully identifies drug candidates that may modulate NOD2 signaling in Crohn's disease. The model:

1. PERFORMS REALISTICALLY
   - AUROC = 0.90 (not inflated by data leakage)
   - Generalizes across chemical scaffolds
   - Passes all validation checks

2. VALIDATES ON KNOWN DRUGS
   - Corticosteroids (standard IBD therapy) rank highly
   - Model captures biologically relevant features

3. IDENTIFIES NOVEL CANDIDATES
   - Antihistamines: Unexpected gut mast cell connection
   - Statins: Epidemiological support for IBD protection
   - Vitamin D: Well-established IBD association
   - Neurosteroids: Novel neuroimmune hypothesis

4. CONNECTS TO TRADITIONAL MEDICINE
   - Bufadienolides (Chan Su) - Chinese anti-inflammatory
   - Bile acids (Niu Huang) - Digestive traditional remedy
   - Validates ethnopharmacological knowledge

FUTURE DIRECTIONS

- Experimental validation of top candidates in cell-based NOD2 assays
- Molecular dynamics simulations of binding poses
- Combination therapy exploration (antihistamine + steroid)
- Natural product isolation and SAR studies